SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

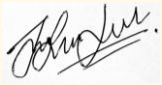
Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – EXPERIMENTATION

Course Code:	ITA03	**Course Title:**	Mobile computing
CO Assessed:	CO2	**Assessment:**	Assessment Tool 2 – EXPERIMENTATION
Weightage:	30 % of CIA	**Total Marks:**	100 marks
CO2 Statement:	*Compare mobile communication technologies, including multiple access techniques and network evolution, based on performance and applications.*		
Student Name:	Joel Benn John B	**Reg. No.:**	1911260003
Section / Batch:	Slot A	**Date:**	17-10-2026

Code of Conduct

I Joel Benn John B (1911260003) (Name / Reg. No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: 

Instructions

- This test contains FIVE questions.
- Duration: 60 minutes (homework)

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Annexure A - QUESTIONS

1. Figma-Based Mobile IP Communication Workflow Analysis (BL4 – Analyze)

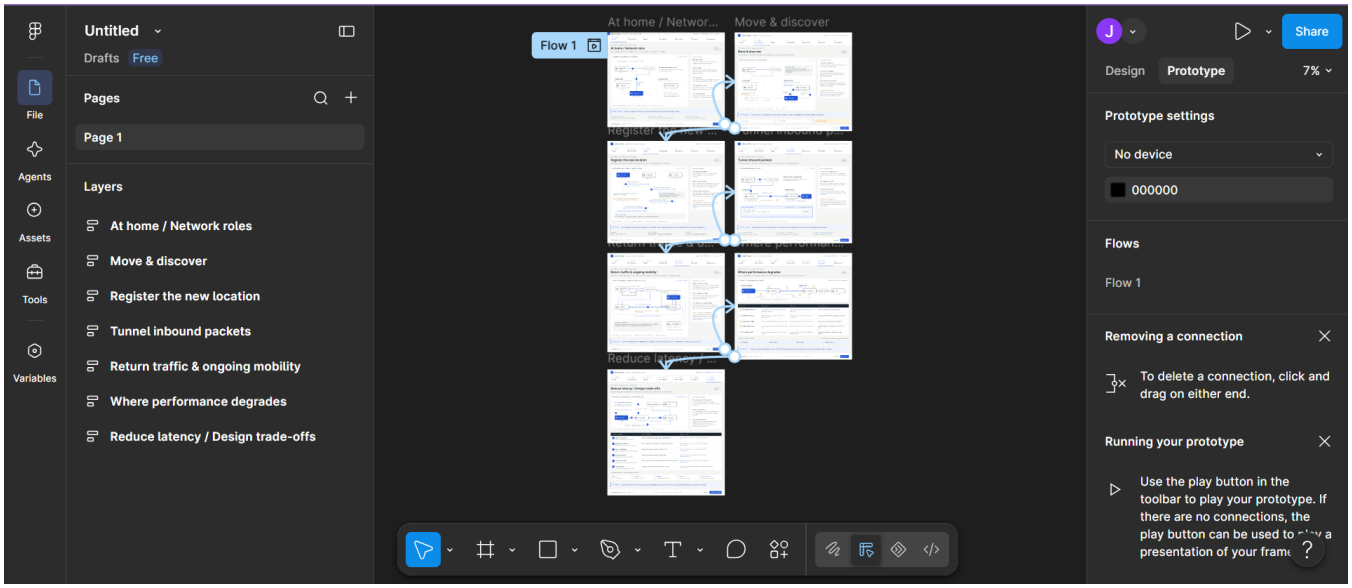
Design an interactive Figma prototype illustrating the complete workflow of Mobile IP communication when a Mobile Node moves from its Home Network to a Foreign Network. Your design should include the Home Agent (HA), Foreign Agent (FA), Correspondent Node (CN), Care-of Address (CoA), Registration Process, and Packet Tunneling.

Analyze the communication process by identifying:

- The purpose of each network component.
- Points where packet delay or packet loss may occur.
- The effect of node mobility on communication performance.
- Possible improvements to reduce latency.

Deliverables:

- Figma (.fig) design
- GitHub repository containing exported images/PDF and documentation (README).



2. Comparative Network Architecture Design Using Figma (BL4 – Analyze)

Using Figma or draw.io, create a comparative architectural diagram showing both a Cellular Wireless Network and a Mobile Ad Hoc Network (MANET) operating in the same smart city environment.

Analyze the architectures by comparing:

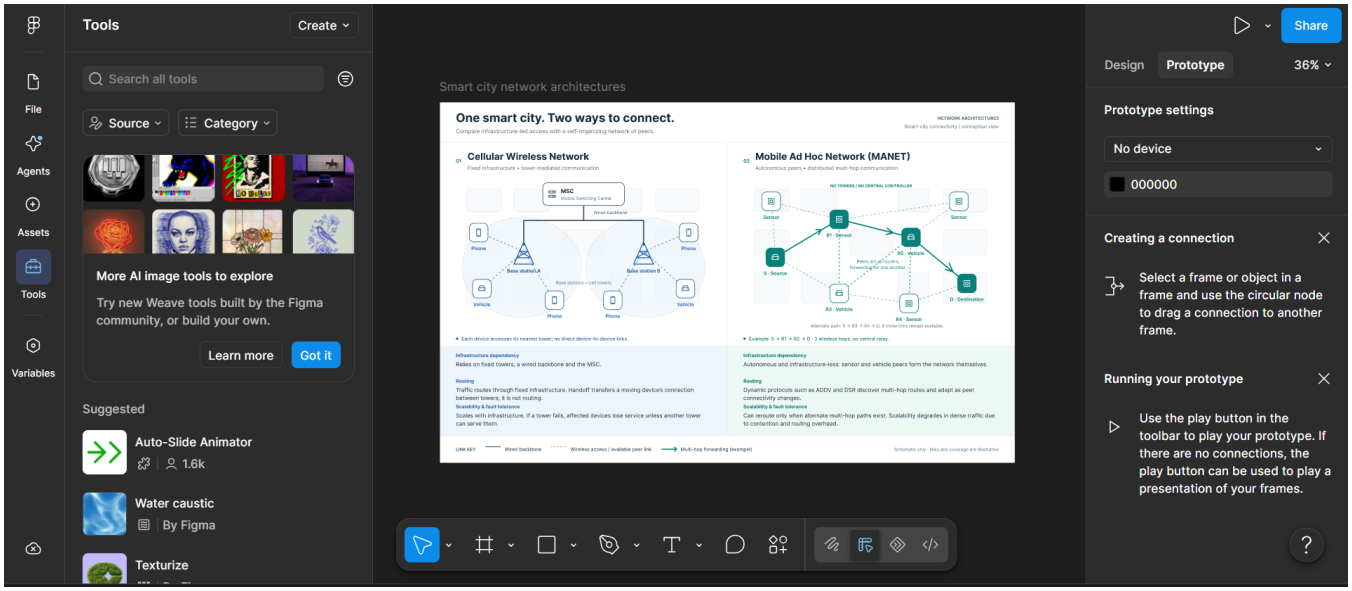
- Infrastructure dependency.
- Routing mechanisms.
- Scalability.

- Fault tolerance.
- Mobility support.
- Suitable application scenarios.

Highlight the advantages and limitations of each architecture through annotations in your design.

Deliverables:

- Editable design file.
- GitHub repository with screenshots, comparison report, and design assets.



3. Routing Protocol Visualization and Analysis (BL4 – Analyze)

Develop an interactive visualization using Figma, Mermaid.js, D3.js, or any visualization framework illustrating the operation of one Proactive Routing Protocol and one Reactive Routing Protocol in a mobile wireless network.

Analyze:

- Route discovery process.
- Route maintenance mechanism.
- Routing overhead.
- Delay in establishing communication.
- Performance under high node mobility.

Include a comparative summary explaining which protocol performs better under different mobility conditions.

Deliverables:

- Source code/design files.
- GitHub repository with implementation, screenshots, and analysis report.

QUESTION 3

MOBILE AD HOC NETWORKS / PROTOCOL ANALYSIS

Routing Protocol Visualization and Analysis

THE CORE DISTINCTION

Maintain routes in advance,
or discover them when needed.

One network. Two approaches to route availability.

S sends to D through relays A and B. Links are bidirectional; only adjacent nodes are in radio range. The topology stays stable during each sequence.



Read timelines ↓
Arrows show messages,
not physical distance.

DSDV

01 / PROACTIVE

Destination-Sequenced Distance-Vector

Routes are ready before data is sent. Every node periodically advertises its routing table to its neighbors.



1. **Exchange tables in the background**
   At t_0, every node broadcasts to all adjacent neighbors. Paired arrows represent advertisements in both directions.
2. **Repeat at each periodic interval**
   At $t_0 + T$ and $t_0 + 2T$, advertisements recur—even during idle periods. Tables retain next hops, hop counts and sequence numbers.
3. **Forward immediately on the known route**
   When data becomes available, S uses its existing route: S → A → B → D. No new route-discovery exchange is required.

Trade-off: low startup delay, but ongoing control traffic.

AODV

02 / REACTIVE

Ad Hoc On-Demand Distance Vector

Discover a route only when data needs to be sent. A request travels out; a reply establishes the path back.



1. **Trigger discovery when data arrives**
   S has data for D but no valid route; it broadcasts a Route Request (RREQ).
2. **Propagate the request to the destination**
   A and B rebroadcast the RREQ; each records a reverse path toward S.
3. **Return the Route Reply (RREP)**
   D unicasts D → B → A → S; relays install forward routes to D.
4. **Send data over the established route**
   After the reply reaches S, data follows S → A → B → D.

Trade-off: less idle control traffic, but a discovery wait.

Comparative summary

BACKGROUND READINESS vs. ON-DEMAND DISCOVERY

Feature	Proactive (DSDV)	Reactive (AODV)
Route Discovery	Periodic / Continuous background updates	On-demand (only when needed)
Routing Overhead	High (constant control traffic)	Low (only during discovery)
Initial Delay	Low (routes are pre-established)	Higher (delay to discover route)
High Mobility Performance	Poor (tables become stale quickly)	Better (adapts to active changes)

TECHNICAL NOTE These are general comparative tendencies. DSDV also uses triggered updates; AODV has route maintenance overhead, and performance depends on mobility and traffic. Timelines are schematic and not to scale.

Key distinction: when control work happens determines the balance between overhead and initial delay.

Q3 / 01

4. Mobile IP Application Design for Smart Healthcare (BL4 – Analyze)

Design a Smart Healthcare Mobile IP system using Figma that enables doctors, ambulances, hospitals, and patients to remain connected while moving across different wireless networks.

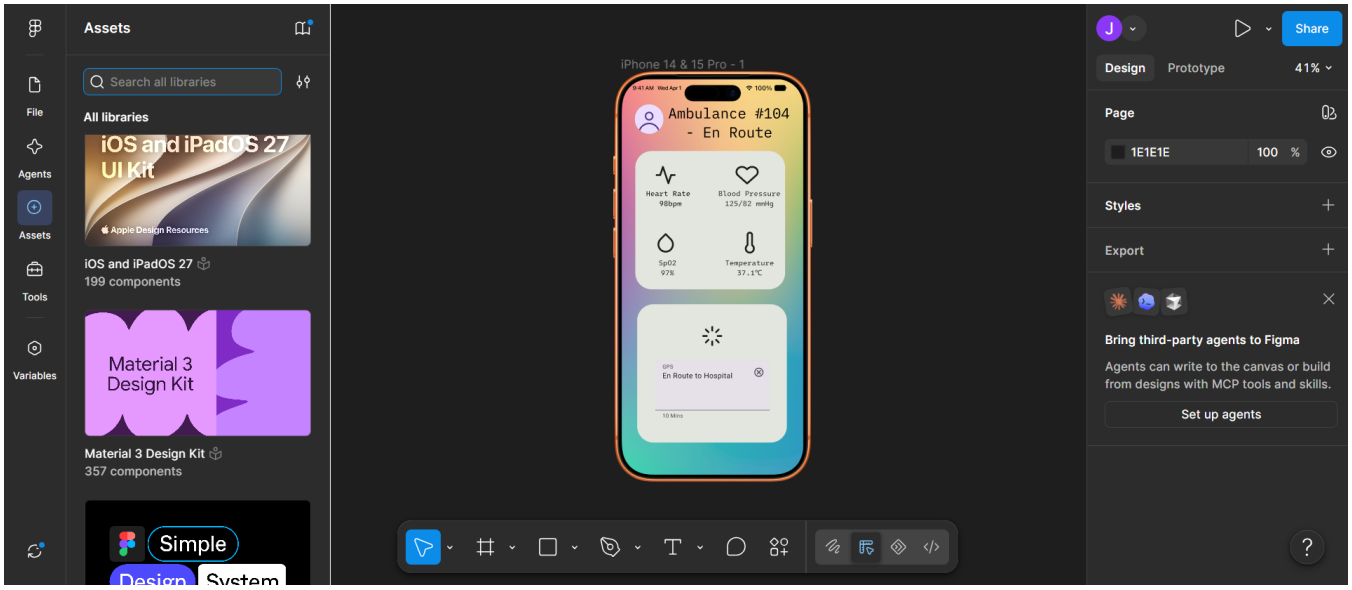
Analyze:

- How Mobile IP maintains uninterrupted communication.
- Challenges during handoff between networks.
- Security vulnerabilities during mobility.
- Possible design improvements to increase reliability.

Provide navigation between screens demonstrating different mobility scenarios.

Deliverables:

- Interactive prototype.
- GitHub repository containing design files, exported prototype, and documentation.



5. Wireless Network Decision Dashboard Using GitHub Project (BL4 – Analyze)

Create a web-based dashboard (using HTML/CSS/JavaScript, React, or another programming framework) or an interactive Figma dashboard that helps network engineers determine whether a given scenario should use Mobile IP, Cellular Networks, or Ad Hoc Networks.

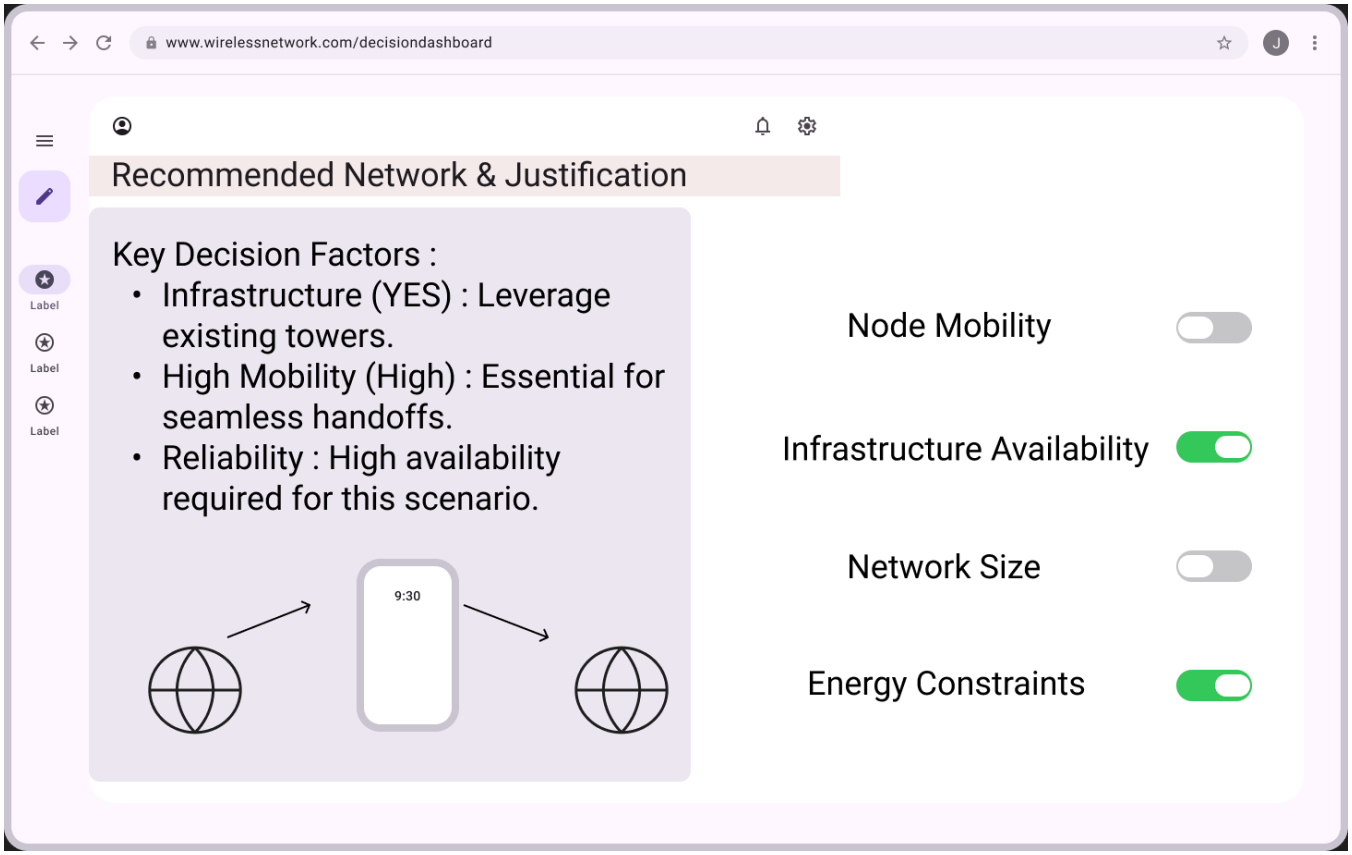
The dashboard should analyze multiple parameters such as:

- Node mobility.
- Infrastructure availability.
- Network size.
- Routing complexity.
- Energy consumption.
- Reliability requirements.

The system should justify its recommendation based on the analyzed parameters.

Deliverables:

- Source code or Figma project.
- GitHub repository with complete code/design, README, screenshots, and analysis report.



Criterion	Marks
Design Quality (Figma/Programming Tool)	20
Technical Analysis	30
Correctness of Mobile Networking Concepts	20
GitHub Repository Organization & Documentation	15
Creativity and Visualization	15
Total	**100 Marks**